



■ CS177 Bioinformatics: Software Development and Applications

Lecture 19: Systems Biology

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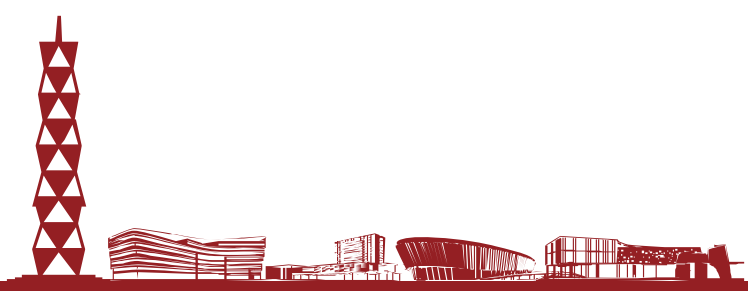
Spring, 2024



Learning objectives



- At the end of this lecture, you will be able to:
 - Explain what is systems biology
 - Give a list of different types of biological networks
 - Analyze the topological characteristics and structures of biological networks
 - Describe the basics of deterministic modeling, e.g. the Michaelis-Menten model
 - Describe the basics of stochastic modeling, e.g. the Gillespie algorithm





Principles of Systems Biology





What is Systems Biology?



- **Systems biology** is concerned with the study of biological functions and mechanisms, underpinning inter- and intra-cellular dynamical networks, by means of system-oriented approaches
- “Life is an emergent, rather than an immanent and inherent, property of matter: Although it arises from the material world, it cannot be reduced to it.” (Erwin Schrödinger, in his classic book “What is Life” , 1944)
- A systems biology approach means
 - Investigating the components of cellular networks and their interactions
 - Applying experimental high-throughput and whole-genome techniques
 - Integrating computational and theoretical methods with experimental efforts





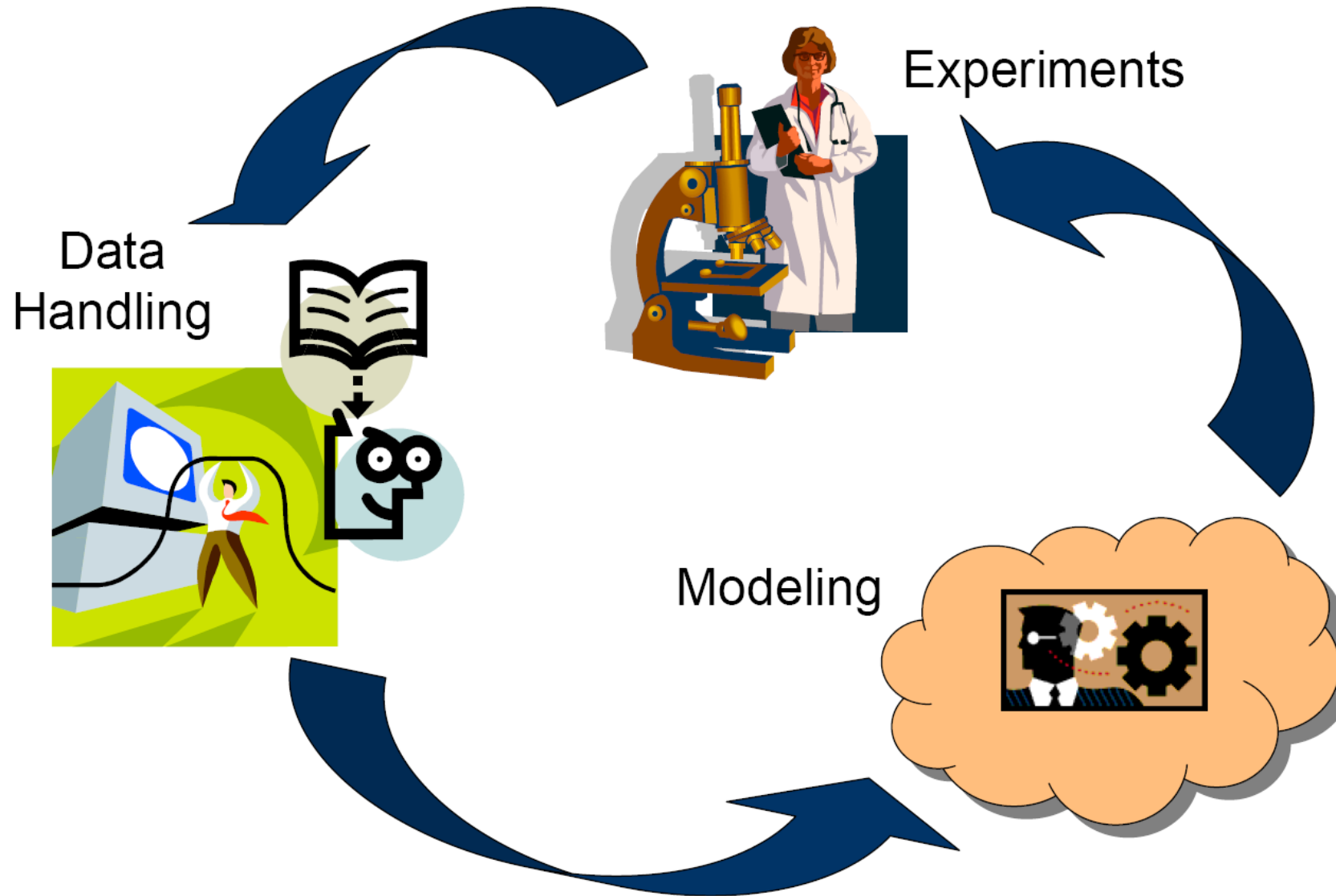
Goals of systems biology



- Construct models to unveil mechanisms causing altered phenotypes and devise novel therapies and drugs
- Predictive tools to design cells with desired properties cheaply and reliably
- Personalized and predictive medicine



An iterative approach





Data integration



- Observation of biological phenomena is restricted to the granularity and precision of the available experimental techniques
- A strong driving force behind the development of a systematic approach in the past 20 years have been given by the new high-throughput biotechnologies
 - Sequencing of human and other genomes (genomics)
 - Monitoring large-scale gene expression levels (transcriptomics)
 - Discovering protein-protein and protein-DNA interactions (proteomics)
- Different types of information need to be integrated





Issues in data integration



- Data representation and storage:
 - (Too) many databases (GO, KEGG, PDB, Reactome ...)
 - XML-like annotation languages (SBML, CellML)
- Information retrieval
 - Tools for retrieving information from multiple remote databases
- Data correlation
 - Find the correlation between phenotypes and genotypes (genomic/proteomic profiles)
 - Statistics, data mining, pattern analysis, clustering, PCA, ...





Models are not real, but reality can be modeled



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- In almost any case, models are only rough representations of their biological counterparts
- Nevertheless, models allow to
 - Elucidate network properties
 - Check the reliability of basic assumptions
 - Uncover the lack of knowledge and requirements for clarification
 - Create large repositories of current knowledge, formalized in a non-ambiguous way and including quantitative data



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What is a model?



- It depends on whom you ask ...
 - **Geneticist:** the mouse family Ts65DN serves as a model for human trisomy 21
 - **Chemist:** a **reaction network**, described by dots (for metabolites) and arrows (for reactions)
 - **Mathematician/Engineer:** the same reaction network can be modeled by a system of nonlinear ODEs (ordinary **differential equations**)
- Abstract representation of objects or processes that explains features of these objects or processes





Overview of Biological Networks





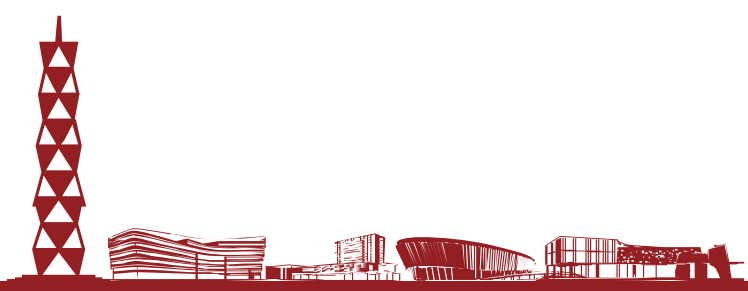
Types of biological networks



- Different kinds of biological networks can be distinguished at the molecular level
 - Gene regulatory network
 - Metabolic network
 - Signal transduction network
 - Protein-protein interaction network
- Moreover, other networks can be considered as we move to different description levels, e.g.
 - Immunological network
 - Neural network
 - Ecological network
- Here, we will focus exclusively on molecular processes that take place **within** the cell



- A major challenge is to **infer** the network of complex macromolecular interactions at the gene, metabolite and protein levels, with reasonable accuracy
- Once identified, the network model can be used to:
 - **Simulate** the process it represents
 - **Predict** the features of its dynamical behavior
 - **Extrapolate** cellular phenotypes

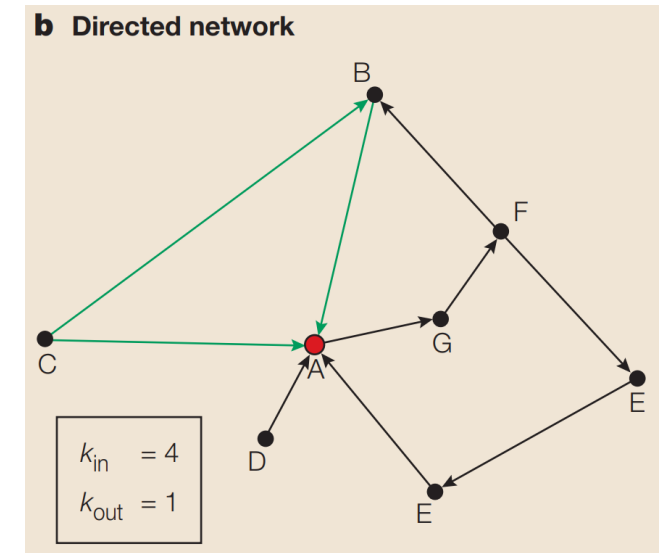
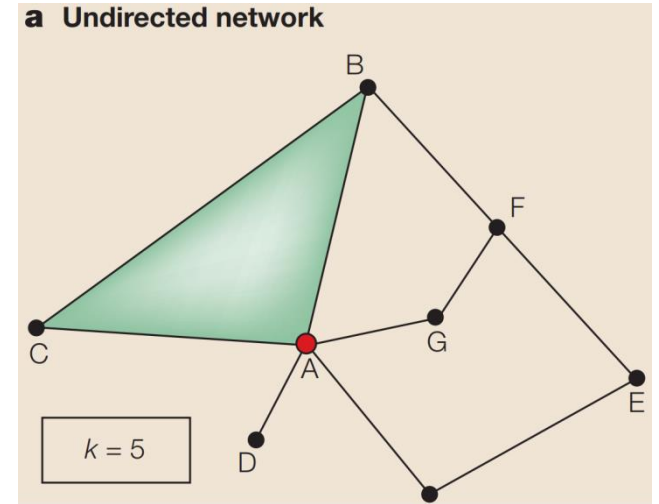




Graphs



- Graph representation is a very useful formal tool for describing and visualizing biological networks
 - An **(undirected) graph** is an ordered pair $G = (V, E)$, where V is the set of the **vertices** (or **nodes**), and E is the set of unordered pairs of distinct vertices, called **edges** (or **lines**)
 - For each edge $\{u, v\}$, the nodes u and v are said to be **adjacent**
 - We have a **directed graph**, or **digraph**, if E is a set of ordered pairs
 - In digraphs, the **in-degree**, k_{in} , (or, **out-degree**, k_{out}) of a node is the number of edges entering into (or, leaving from) that node

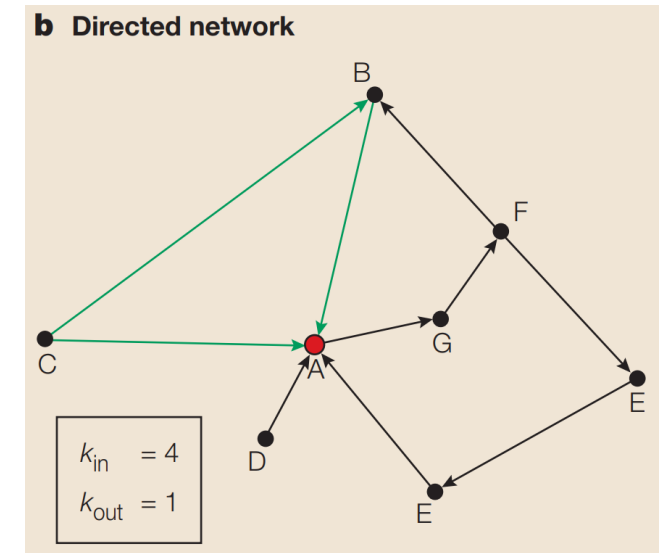
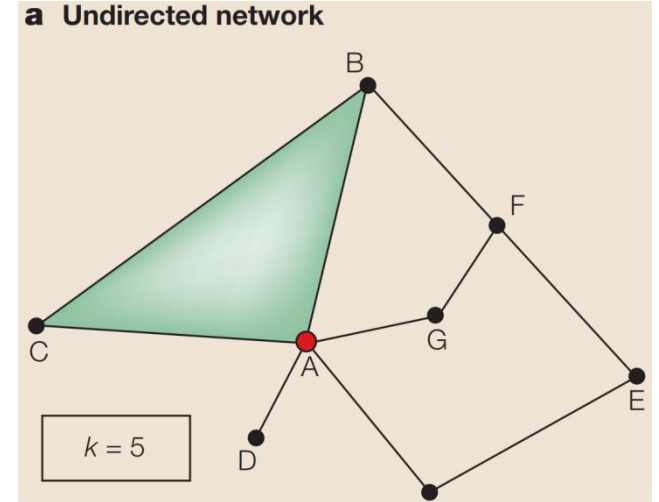


Barabasi et al. Nature Review
Genetics, 101(5), 101-113, 2004

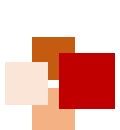


Topological characteristics

- The **degree distribution**, $P(k)$, gives the probability that a selected node has exactly k links
- It allows us to distinguish between different classes of networks (see next slide)
- The **clustering coefficient** of a node I , C_I , measures the aggregation of its adjacencies (i.e. the number of triangles passing through node I)
- $C(k)$ is the average clustering coefficient of all nodes with k links



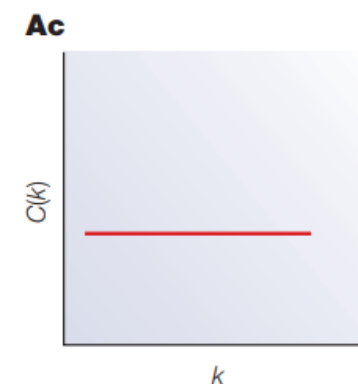
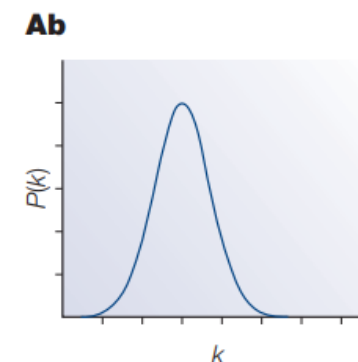
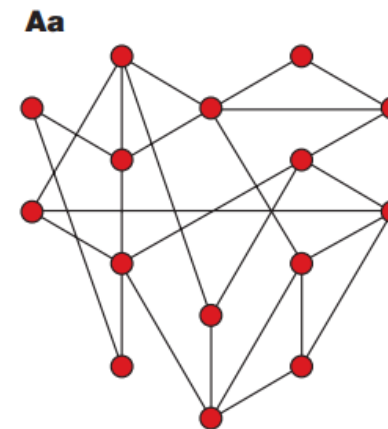
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Erdős-Rényi random networks



- The Erdős-Rényi model of a random network starts with N nodes and connects each pair of nodes with probability p
- The degree allows a **Poisson** distribution, thus many nodes have the same number of links (close to the average degree $\langle k \rangle$)
- The tail decreases exponentially, which indicates that nodes with degree k very different from the average degree are rare
- The clustering coefficient is independent of a node's degree



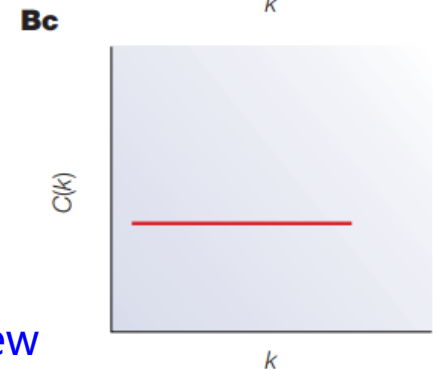
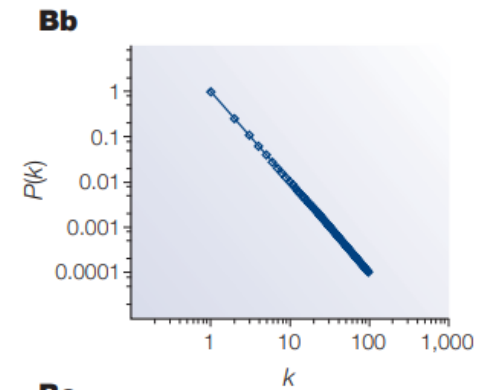
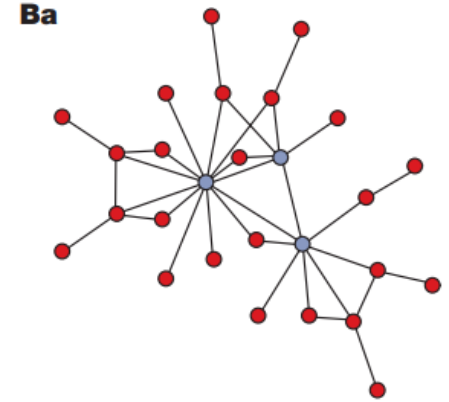
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Scale-free networks



- Scale-free networks are characterized by a **power-law degree distribution**
- The probability that a node has k links follows $P(k) \sim k^{-\gamma}$, where γ is the degree exponent
- The probability that a node is highly connected is statistically more significant than in a random graph
- In the **Barabási-Albert model**, at each time point a node with M links is added to the network, which connects to an already existing node i with probability $\Pi_i = k_i / \sum_j k_j$
- The underlying mechanism is that nodes with many links have higher probability of getting more (this is also referred to as **preferential attachment**)

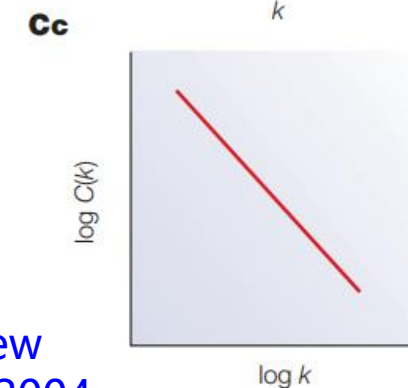
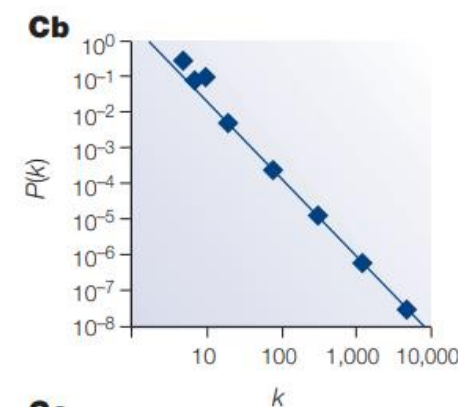
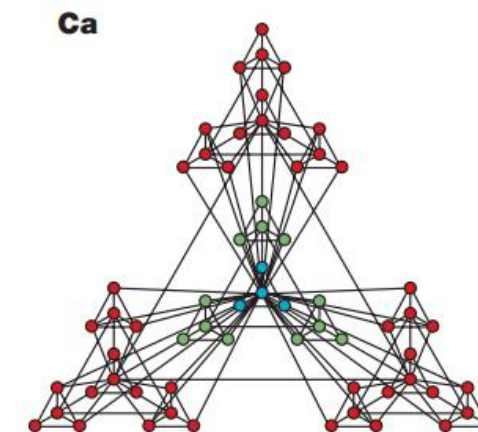


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Hierarchical networks

- A hierarchical structure arises in systems that combine modularity and scale-free topology
- The hierarchical model is based on the replication of a small cluster of four nodes (the central ones)
- The external nodes of the replicates are linked to the central node of the original cluster
- The resulting network has a power-law degree distribution, thus it is scale-free
- The average clustering coefficient scales with the degree, following $C(k) \sim k^{-1}$



Barabasi et al. Nature Review
Genetics, 101(5), 101-113, 2004





Topology of biological networks



- An extensive commentary has been published by Albert in 2005, reviewing the literature on the topology of different kinds of biological networks

Réka Albert, Scale free networks in cell biology, *Journal of Cell Science*, 118(21), 4947-4957, 2005

- Experimental evidences are reviewed for metabolic, transcriptional regulatory, signal transduction, and functional association networks
- All of the considered networks approximately exhibit power-law degree distribution, at least for the in-degree or out-degree
- For instance, transcriptional regulation networks
 - Exhibit a scale-free out-degree distribution, signifying the potential of transcription factors (TFs) to regulate multiple targets
 - On the other hand, their in-degree distribution is a more restricted exponential function, suggesting that combinatorial regulation by several TFs is less frequent



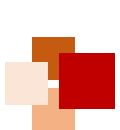


Biological meanings of nodes and edges



- Depending on the kind of biological network, the edges and nodes of the graph have different meanings
- Metabolic network
 - nodes: metabolic product, edge: a reaction transforming A into B
- Transcriptional regulation network (protein-DNA interaction)
 - nodes: genes and proteins, edge: a TF regulates a gene
- Protein-protein interaction network
 - nodes: proteins, edge: interaction between proteins
- Gene regulatory networks (functional association between genes)
 - nodes: genes, edge: expressions of A and B are correlated



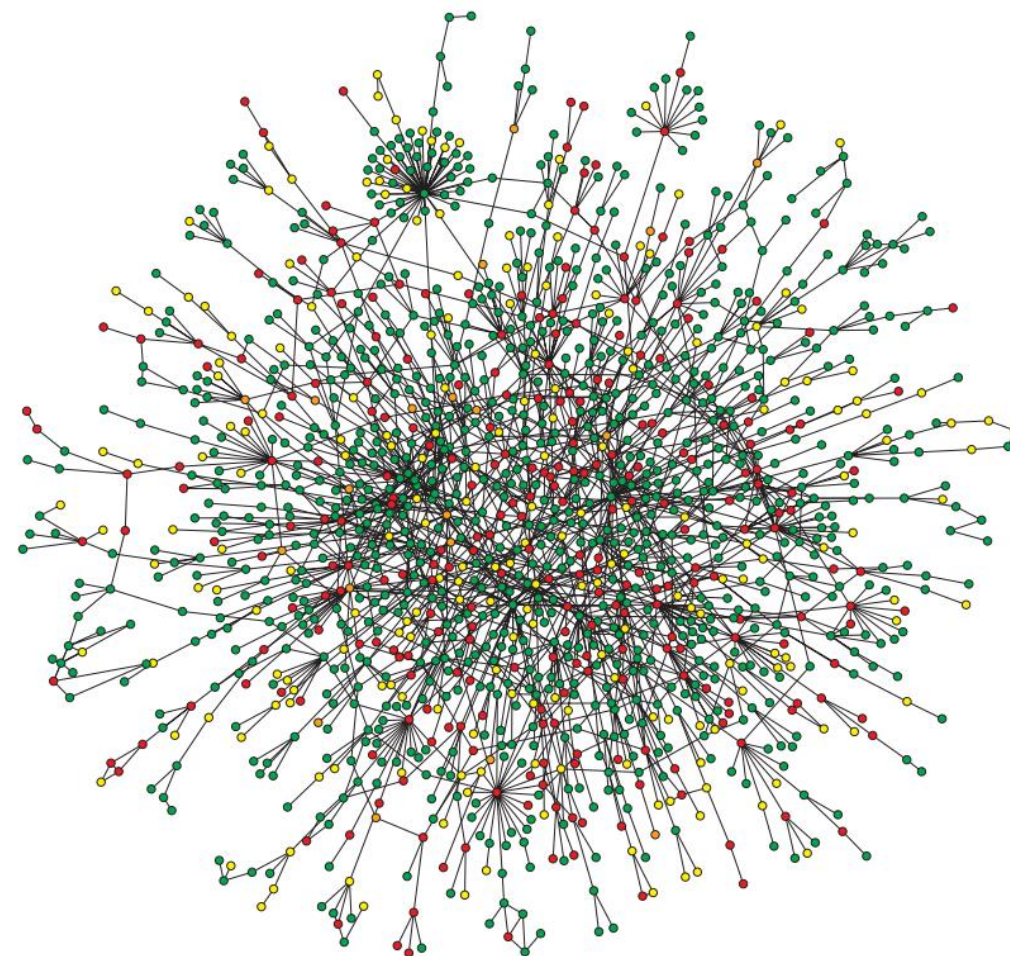


Protein-protein interaction network



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- This network is based on **yeast two-hybrid (Y2H) experiments**
- A few highly connected nodes (hubs) hold the network together
- The color of a node indicates the phenotypic effect derived from removing the corresponding protein
 - **Red**: lethal
 - **Green**: non lethal
 - **Orange**: slow growth
 - **Yellow**: unknown



Barabasi et al. Nature Review
Genetics, 101(5), 101-113, 2004



Metabolic reactions



- Metabolic reactions can be divided into two categories
 - **Catabolic reactions**: breakdown of complex compounds to get energy and building blocks
 - **Anabolic reactions**: assembling of the compounds used by the cellular mechanisms
- Living cells require energy and material for:
 - Building up membranes
 - Storing molecules
 - Replenishing enzymes
 - Replication and repair of DNA
 - Movement





Basic concepts of metabolism



- Historically, metabolism is the part of cell functioning that has been studied more thoroughly than other cellular networks during the past few decades
- This implies that several well-studied mathematical tools exist for describing metabolic networks:
 - **Enzyme kinetics** investigates the dynamic properties of the individual reactions in isolation
 - **Stoichiometric analysis** deals with the balance of compound production and degradation at the network level
 - **Metabolic control analysis** describes the effect of perturbations in the network, in terms of changes of metabolites' connections
- Most of the tools used in the quantitative study of metabolic networks can also be applied to other types of networks





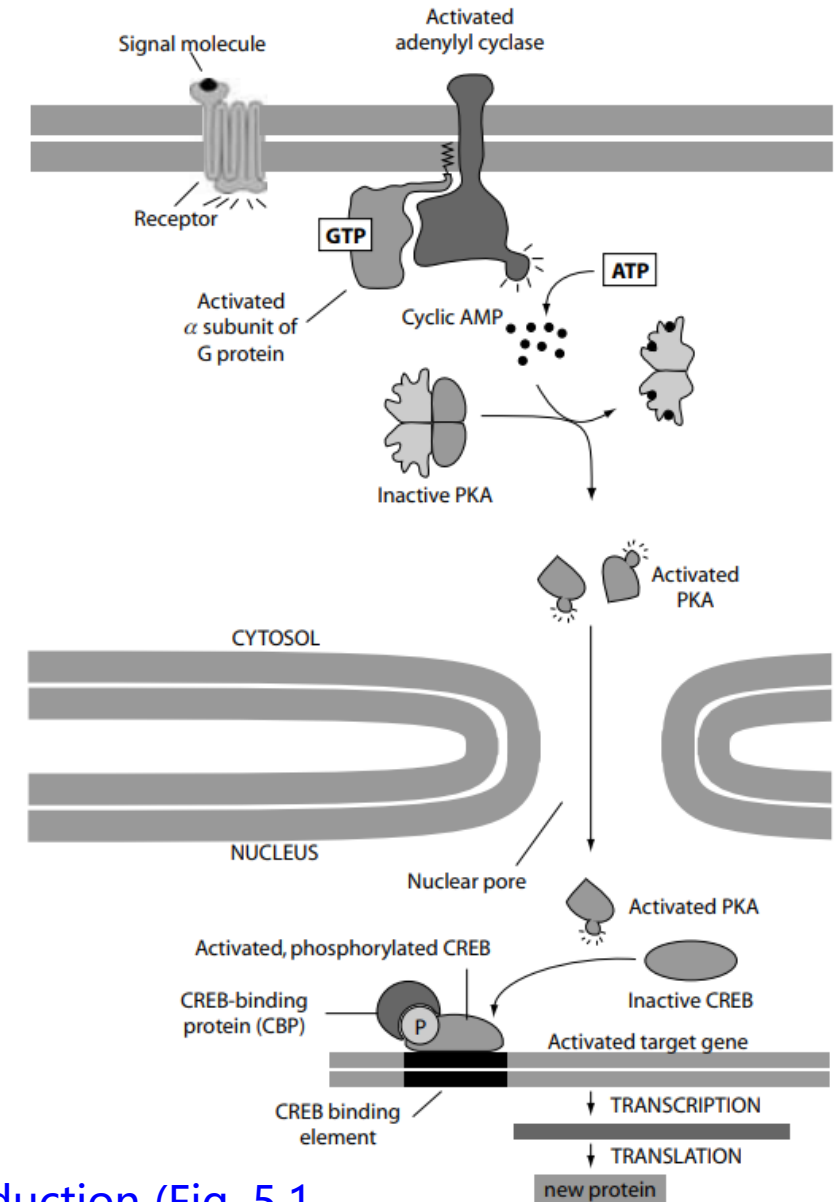
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B. Palsson, *Systems Biology: Properties of Reconstructed Networks*, Cambridge University Press, 2006

Signaling network

- Signaling networks (pathways) involve the transduction of a “signal” from the outside to the inside of a cell
- Three fundamental modes of signal transduction:
 - One cell sends a soluble signal that diffuses to the target cell
 - Cells manipulate the composition of the extracellular matrix
 - Cells communicate through specific and direct cell-to-cell mechanisms
- Comparison with metabolism:
 - Metabolism involves breakdown of molecules for energy production and synthesis of metabolites
 - Signaling conveys “information”, often leading to the change of transcription state of the genome

A 3-level view of signal transduction (Fig. 5.1, Palsson's book 2006)





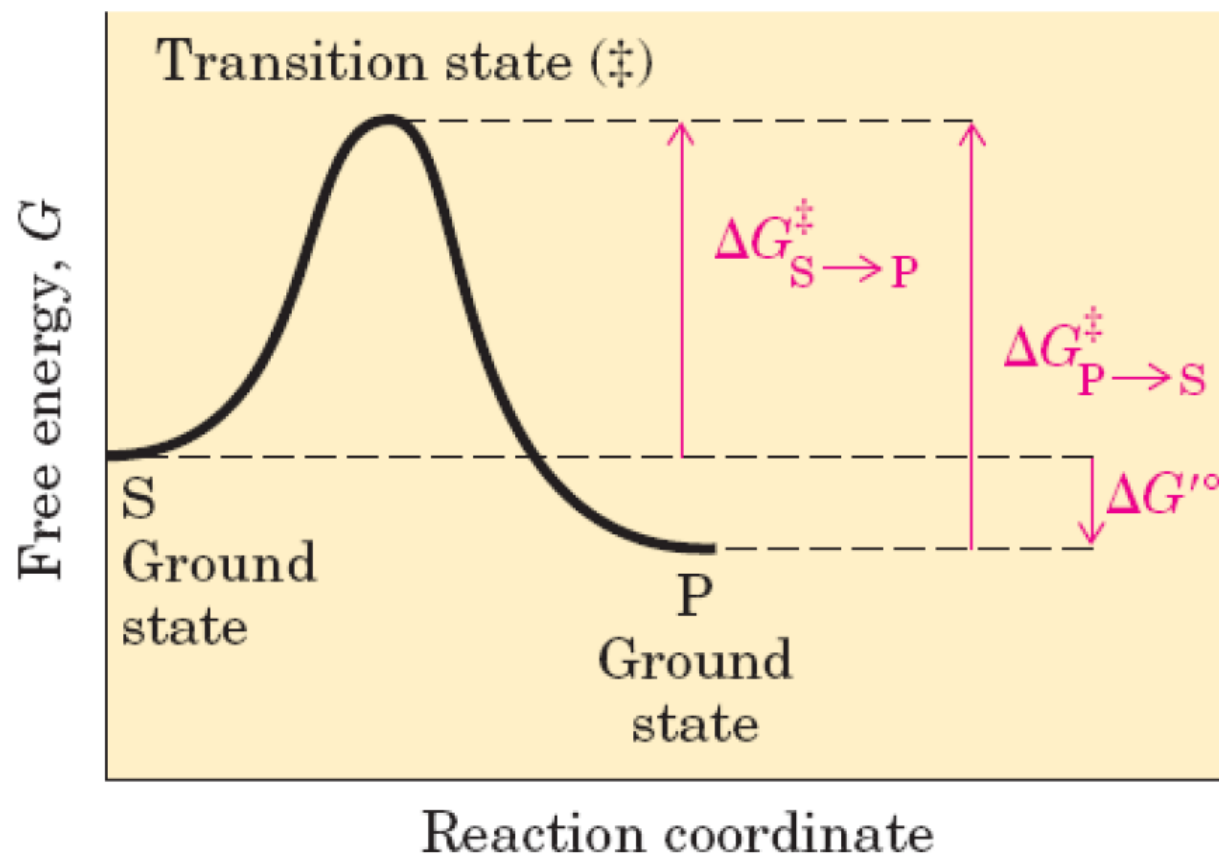
Deterministic Modeling of Biochemical Reactions

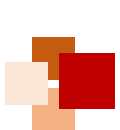


Biochemical energy



- The **equilibrium** of a reaction is linked to the variation of biochemical standard **free energy**, $\Delta G'^{\circ}$
- The **velocity**, instead, depends on the **activation energy**, $\Delta G_{S \rightarrow P}^{\ddagger}$





Reaction rate



- The **reaction rate** is determined by
 - The **concentration** of the reactants
 - The **kinetic constant**, usually denoted by k
- The reaction rate for $S \xrightarrow{k} P$ is

$$V = k[S]$$

- From **transition-state theory (TST)**, it is possible to derive the relation

$$k = \frac{kT}{h} e^{-\Delta G^\ddagger/RT}$$

k : Boltzmann constant

h : Planck constant

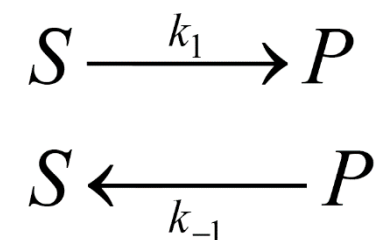


Reaction equilibrium



- In a chemical reaction, **chemical equilibrium** is the state in which both reactants and products are present in concentrations which have no further tendency to change with time

- Let us consider the following basic reversible reaction



- It can be described by the system of ODEs below:

$$\begin{aligned} \frac{d[S]}{dt} &= -k_1[S] + k_{-1}[P] \\ \frac{d[P]}{dt} &= k_1[S] - k_{-1}[P] \end{aligned}$$

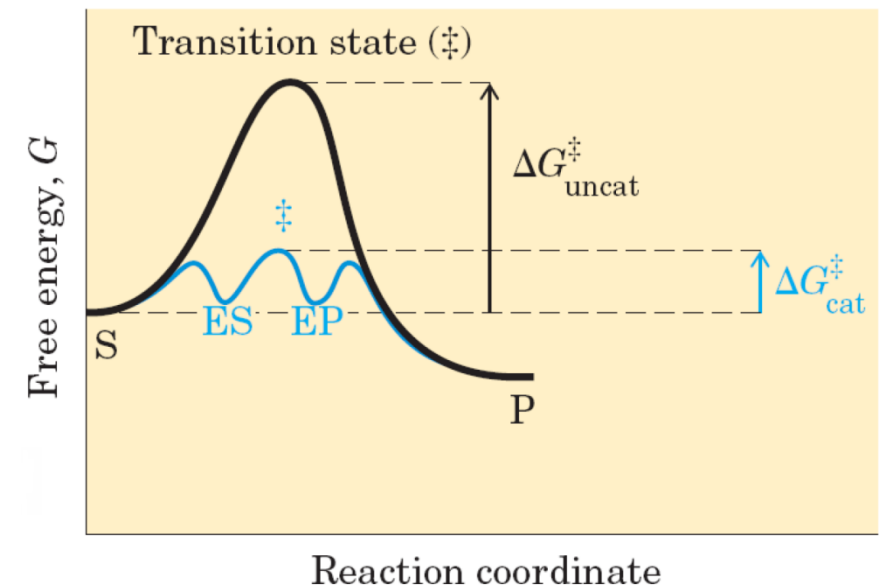


$$K'_{eq} = \frac{[P]}{[S]} = \frac{k_1}{k_{-1}}$$



Enzymatic reaction kinetics

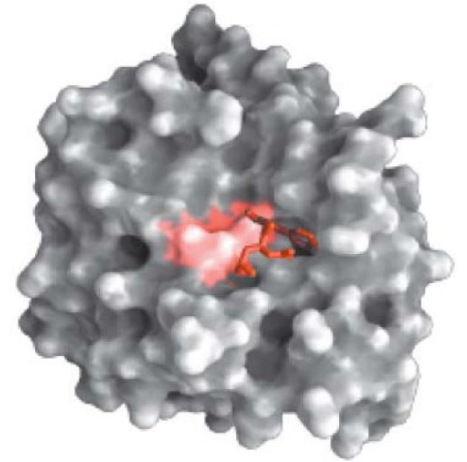
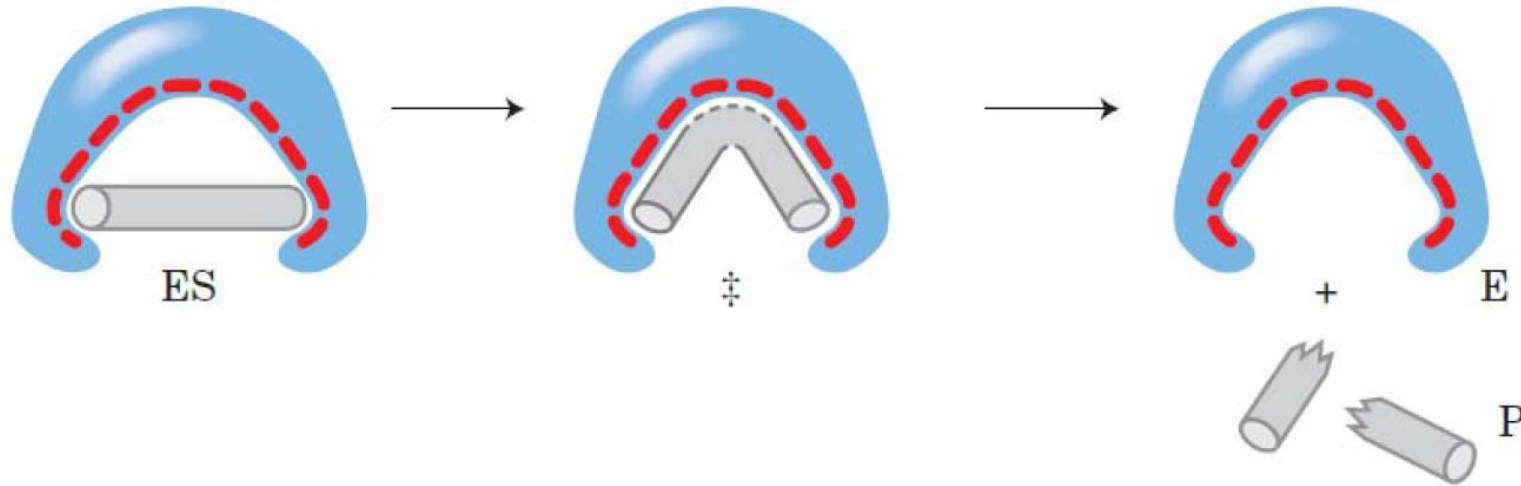
- **Enzymes** are a family of proteins specialized in the catalysis of reactions
- Catalysts do not react and do not affect the reaction equilibrium. However, they increase the reaction rate by decreasing the activation energy
- The basic mechanism is the formation of an enzyme-substrate complex, which creates a more favorable condition for the formation of the product
- After the substrate has been transformed into product, the complex dissociates and the enzyme is free to catalyze the next reaction

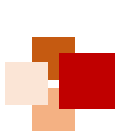


The role of enzymes



- Enzymes play a central role in all the biological processes (e.g. metabolism, regulation, signaling)
- Many diseases are caused by deficiency of some enzyme
- Many drugs act by interacting with enzymes

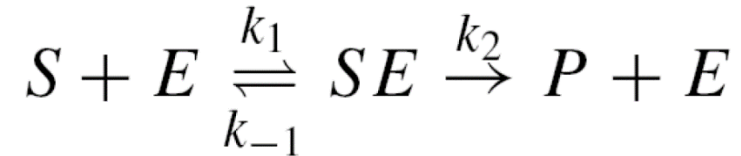




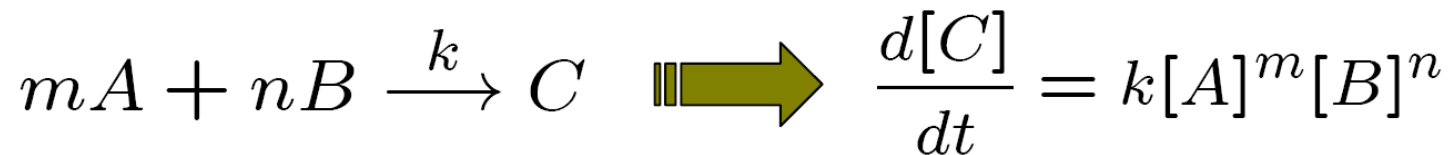
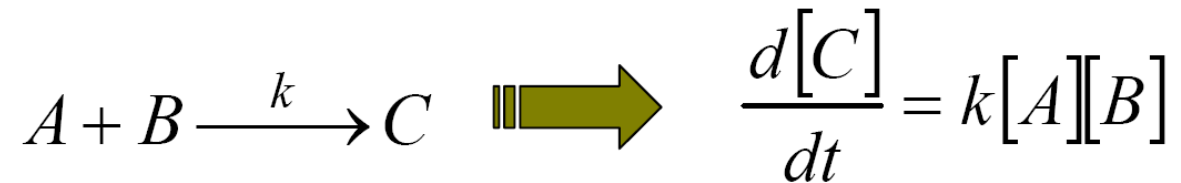
Michaelis-Menten model



- The basic model of enzymatic reaction was proposed by Michaelis and Menten in 1913



- The **law of mass action (LMA)** states that the reaction rate is proportional to the product of the concentrations of the reactants



Leonor Michaelis
(1875 - 1949)



Maud Menten
(1879 - 1960)



Michaelis-Menten model (cont' d)

- Applying the LMA to the model reaction scheme, we obtain

$$\begin{aligned}\dot{s} &= -k_1 es + k_{-1}c, & \dot{e} &= -k_1 es + (k_{-1} + k_2)c \\ \dot{c} &= k_1 es - (k_{-1} + k_2)c, & \dot{p} &= k_2 c\end{aligned}$$

$$s := [S], e := [E], c := [ES], p := [P]$$

with initial conditions: $s(0) = s_0$, $e(0) = e_0$, $c(0) = 0$, $p(0) = 0$

- In the above equations, $\dot{x}$ means the derivative of variable x with respect to time (Newton's notation)
- Note: The equation of $p(t)$ is decoupled and yields

$$p(t) = k_2 \int_0^t c(\tau) d\tau$$



Simplified M-M model



- The total amount of enzyme (free + bound) remains constant over time, i.e. summing the equations of $e(t)$ and $c(t)$ equals a constant

$$\dot{e} + \dot{c} = 0 \quad \Rightarrow \quad e(t) + c(t) = e_0$$

- From the above, we can derive $e(t)$ and substitute into the other equations, obtaining

$$\dot{s} = -k_1 e_0 s + (k_1 s + k_{-1})c, \quad s(0) = s_0$$

$$\dot{c} = k_1 e_0 s - (k_1 s + k_{-1} + k_2)c, \quad c(0) = 0$$





Simplified M-M model (cont' d)

- Typically, the initial formation of the complex ES is much faster than the product formation, hence it can be assumed to be instantaneous
- This assumption lets us pose $dc/dt \cong 0$, that is

$$c(t) = \frac{e_0 s(t)}{s(t) + K_m}$$

where the positive constant

$$K_m = \frac{k_{-1} + k_2}{k_1}$$

is the **Michaelis-Menten constant**

- Furthermore, the amount of enzyme is typically much less than that of substrate (at least in metabolic reactions), so one can assume that $s(0) = s_0$ also in the simplified model



Cooperative reactions



- In the basic model of enzymatic reaction, we have assumed that each molecule of enzyme only binds one substrate molecule
- However, it is quite common to have multiple binding sites, e.g. the hemoglobin has 4 binding sites for oxygen
- An enzymatic reaction is said to be **cooperative** if the binding of one molecule to one site affects the binding affinity at other sites
- Such reactions are often described with **Hill equations**





Stochastic Modeling

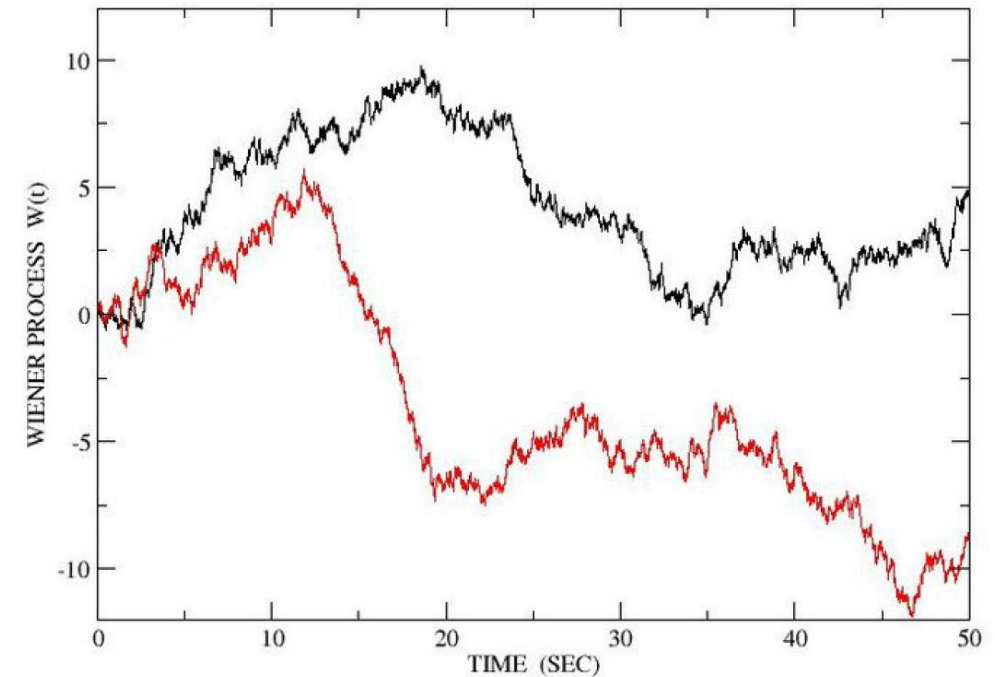




Stochastic models



- The models considered so far are **deterministic**, i.e. given the initial conditions (the state at t_0) and the exogenous signals perturbing the model, the response is univocally determined
- A **stochastic** model, on the contrary, involves **random variables**, therefore its behavior cannot be predicted *a priori*, although it can be statistically characterized
- The figure shows two realizations of the same stochastic process, starting from the same initial condition





Biological systems are stochastic



- Certainly biological systems fall in the category of stochastic systems, because the very basic steps of every molecular reaction can be described **only** in terms of its **probability of occurrence**
- Moreover, the diffusion of molecules is a realization of a random walk process (**Brownian motion**)
- So, why are deterministic models still so widespread?
 - Usually, the phenomena under consideration involve **a large number of molecules**, therefore the average effect is well described through deterministic equations
- When is it necessary to use stochastic models/simulations?
 - When the mechanism to be described is based on the interaction of **few molecules**, or we want to simulate the functioning of a little pool of cells





Stochastic vs. deterministic constants



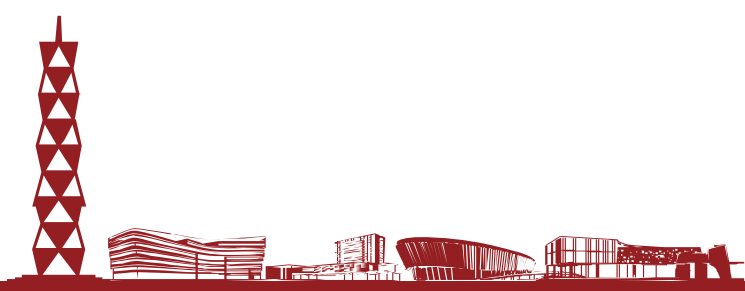
- Intuitively, the **stochastic reaction constant**, c_μ , must be linked to the **kinetic constant** of the corresponding deterministic equation, k_μ
- In the validity ranges of the deterministic models, it is possible to establish the simple relation

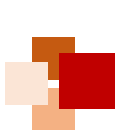
$$k_1 = Vc_1$$

- The presence of the factor V (volume) depends on the fact that deterministic models take as state variables the **concentrations**, whereas stochastic ones use **the number of molecules**



- There are two main approaches to solving the stochastic system and compute the time evolution of the system
 - Master Equation Approach
 - Stochastic Simulation Algorithm (SSA) with Gillespie method





Master equation approach



- The key element in this approach is the probability function

$$P(X_1, X_2, \dots, X_N; t)$$

Probability that at time t
there are X_1 molecules of
species S_1 , ..., X_N of species S_N

- The master equation describes the evolution of the function P
- The probability $P(X_1, \dots, X_N; t + dt)$ can be derived as a combination of the probability of all the possible reactions that can happen within dt
- The differential equation as derived by this argument usually does not have an analytical solution, nor does it admit a computationally efficient solution



Gillespie algorithm



- The algorithm was first presented in
 - D. T. Gillespie (1976). A general method for numerically simulating the stochastic time evolution of coupled chemical reactions.
Journal of Computational Physics, 22: 403-434.
- The purpose is to simulate chemical reactions with limited computational resources
- The outcome is a realization of the underlying stochastic process





Basic principles of Gillespie algorithm



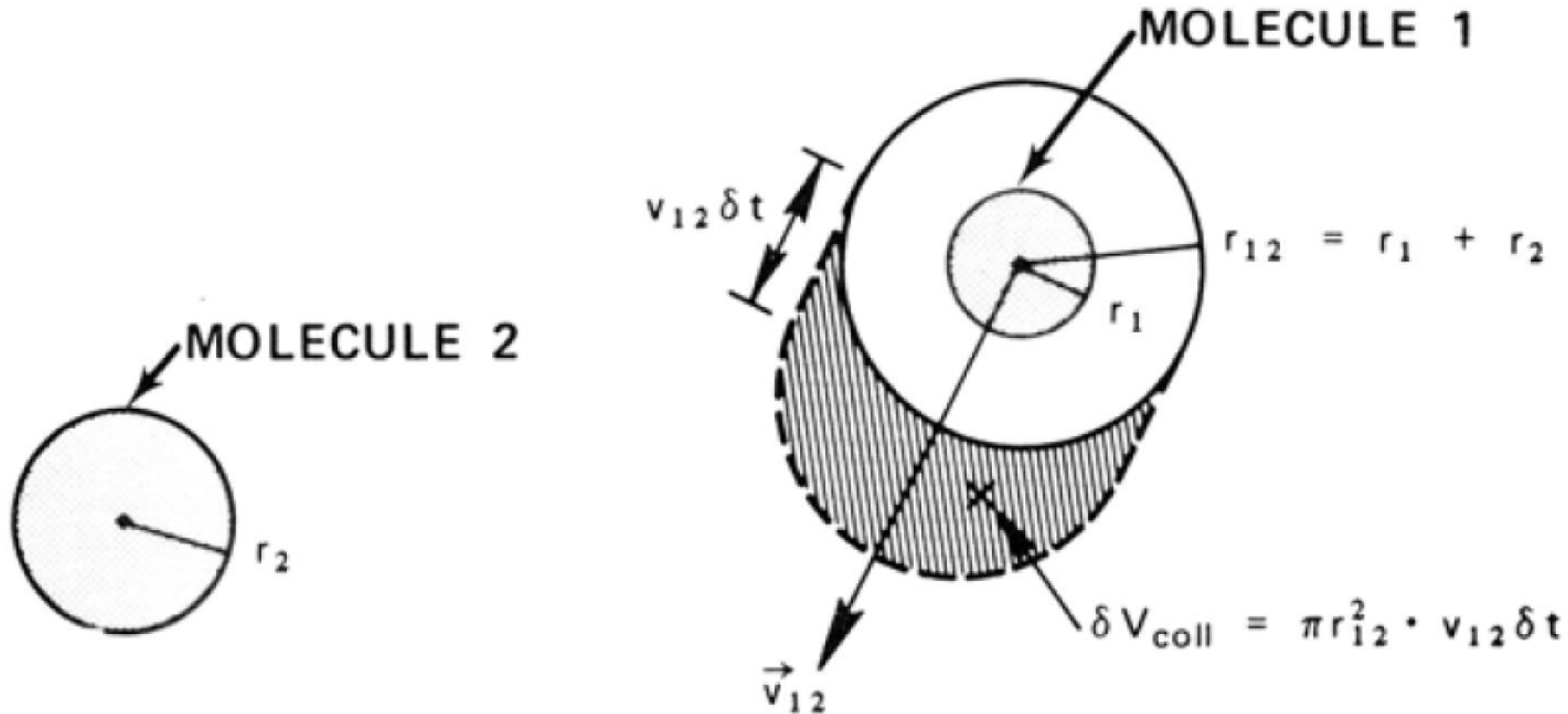
- The basic working principles of the Gillespie algorithm are derived from the description of the collision of particles in a vessel
- When two molecules collide, the reaction happens only if they have proper orientation and kinetic energy
- The first hypothesis is that the frequency of non-reacting collision is much larger than that of reacting ones
- Secondly, the algorithm assumes that each reaction involves no more than two molecules



■ Sketch of the derivation



- The method starts from the observation of the volume occupied by **molecule 1** that is in relative motion with respect to another one, **molecule 2**





Sketch of the derivation (cont' d)



- Assuming a random uniform distribution of the molecules in the volume V , the **probability of collision** can be computed as

$$\delta V_{\text{coll}}/V = V^{-1} \pi r_{12}^2 \bar{v}_{12} \delta t$$

Probability of collision of two molecules in the time interval δt

- If in such a volume we have X_1 molecules of the species S_1 and X_2 of the species S_2 , the probability of their collision is

$$X_1 X_2 V^{-1} \pi r_{12}^2 \bar{v}_{12} \delta t$$



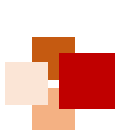


Steps of the Gillespie algorithm



- **Step 0 (Initialization)**: Define the number of molecules of each species, the kinetic constants and the random number generator
- **Step 1 (Monte Carlo)**: Generate random numbers, to determine which is the next reaction and the length of the time interval dt
- **Step 2 (Update)**: Increase time by dt and update the number of molecules of each species on the basis of the occurred reaction
- **Step 3 (Iterate)**: If the number of reactants is greater than zero and the simulation stop time has not yet been reached, iterate from Step 1

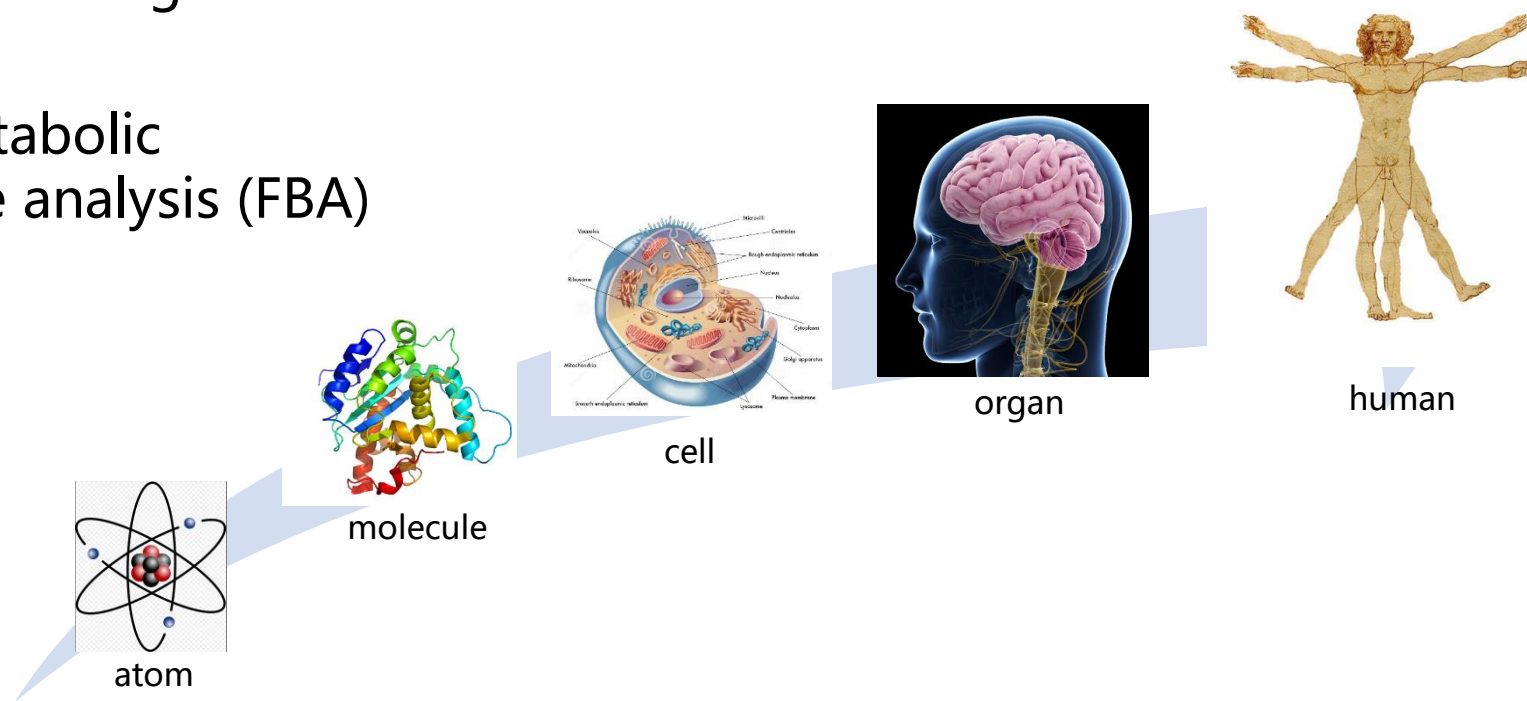




Other topics of systems biology



- Integration of multi-modal and multi-omics data (genomics, transcriptomics, proteomics, epigenomics, etc.)
- Inference of biological networks from omics data, e.g. inference of gene regulatory network (GRN) from single-cell gene expression data
- Simulate and predict the metabolic phenotypes, e.g. flux balance analysis (FBA)
- Whole-cell modeling
- Virtual physiological human
- Systems medicine





Recap



- In this lecture, we have learned:
 - Basic principles and challenges of systems biology (e.g. construct predictive models, data integration)
 - Different types of biological networks (metabolic network, signaling network, protein-protein interaction network, gene regulatory network, etc.)
 - Topological characteristics (e.g. degree distribution, clustering coefficient) and structures of biological networks (e.g. scale-free, hierarchical)
 - Basics of deterministic modeling, e.g. Michaelis-Menten model
 - Basics of stochastic modeling, esp. the Gillespie algorithm





References



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- Daniel T. Gillespie (1976) "A general method for numerically simulating the stochastic time evolution of coupled chemical reactions" , *Journal of Computational Physics*, 22: 403–434. (**The Gillespie Algorithm**)

